

Carelix: An AI-Driven Collaborative Multidisciplinary Care and Longitudinal Progress Monitoring Framework for Autism Spectrum Disorder

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Abstract: Autism Spectrum Disorder (ASD) necessitates continuous, collaborative intervention across home, school, and clinical environments. However, conventional monitoring workflows are hindered by fragmented communication, observational subjectivity, and delayed detection of behavioral regressions. This paper introduces **Carelix**, an intelligent digital health ecosystem designed to unify longitudinal progress monitoring for children with ASD. Carelix synchronizes tri-party weekly evaluations from parents, special educators, and clinicians, coupled with standardized M-CHAT-R/F screening. An integrated dual-stage Machine Learning (ML) pipeline extracts cross-environment behavioral discrepancies and classifies developmental trajectories to identify regression risks in advance. Concurrently, a Large Language Model (LLM) reasoning module transforms identified domain weaknesses into tailored, actionable classroom interventions and home routines. Furthermore, a decentralized Ethereum smart contract layer guarantees verifiable, parent-controlled data governance. Experimental evaluation across 600 longitudinal observational records shows that the engineered model achieves a 5-fold cross-validation accuracy of $71.1\% \pm 4.4\%$ (with peak folds reaching 77.3%) and a recall of 0.73 for identifying positive trajectory shifts. Carelix provides a secure, proactive, and data-driven paradigm for multidisciplinary pediatric developmental care.

Keywords: Autism Spectrum Disorder, Machine Learning, Multimodal Telemetry, Large Language Models, Smart Contracts, Pediatric Healthcare.

Introduction: Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition characterized by persistent difficulties in reciprocal social communication, sensory dysregulation, and restricted or repetitive behavioral patterns [1]. Effective therapeutic interventions require sustained, multidisciplinary coordination between caregivers, educators, and clinicians [2]. Despite clinical consensus on early and continuous intervention [3], contemporary workflows suffer from severe structural fragmentation. Children exhibit context-dependent behaviors; a child may show structured compliance in class while experiencing severe sensory overload at home. Because stakeholders record observations in isolated silos, clinicians routinely rely on retrospective anecdotal recall during infrequent consultations [4]. Furthermore, cross-institutional exchange of sensitive pediatric behavioral records introduces substantial data sovereignty and privacy compliance barriers [5]. Existing assistive applications fail to algorithmically triangulate multi-observer metrics into predictive developmental trajectories and actionable, automated intervention guidance. Carelix addresses these challenges by integrating real-time tri-party synchronization, predictive machine learning, generative clinical recommendation synthesis, and decentralized consent management.

Objective (s): The primary objectives of this study are: (1) to develop a synchronized tri-party telemetry platform capturing multi-environment quantitative and qualitative behavioral observations from parents, educators, and clinicians; (2) to design a dual-stage predictive Machine Learning pipeline that models cross-observer discrepancy gaps (Δ_{env}) to classify regression risks and forecast score changes; (3) to construct a Generative AI clinical guidance engine using Large Language Models (LLMs) to automatically translate skill deficits into domain-specific classroom strategies and home coaching routines; and (4) to implement a decentralized smart-contract consent protocol providing immutable, parent-governed data access control.

Methodology: The Carelix framework comprises four interconnected layers:

1. Tri-Party Synchronization & Feature Engineering: Tailored client interfaces collect structured weekly 5-point Likert ratings across four domains: Communication (Q1, Q2), Social Interaction (Q3, Q4), Emotional Regulation & Flexibility (Q5, Q6), and Home Adaptive Living (P1, P2, P3), alongside quantitative metrics (meltdowns, sleep, appetite) and digitalized M-CHAT-R/F screenings. Longitudinal records from a cohort of 50 pediatric subjects across 12-week observational cycles (600 total observations) were engineered to extract: the Cross-Setting Discrepancy Gap ($\Delta_{\text{env}} = |\mu_{\text{parent}} - \mu_{\text{educator}}|$), Behavioral Dispersion ($\sigma_{\text{consistency}} = [\Sigma(x_i - \bar{x})^2 / N]^{1/2}$), Historical Baselines (prev_week_score), and Weakest Domain Localization ($\text{Area}_{\text{weak}} = \text{argmin}_k[\text{Area}_k]$).

2. Dual-Stage Predictive Machine Learning: The feature space (30 engineered features) was divided into training (N=440) and test (N=110) sets with balanced class distributions. A balanced Random Forest classifier (200 estimators, max depth 8) estimates trajectory improvement probabilities and risk categorizations, while a Gradient Boosting regressor estimates expected rate of score delta.

3. Generative AI Pedagogical & Behavioral Guidance: When skill deficits are flagged, an LLM reasoning engine (Meta Llama-3.1-8B-Instant via Groq) applies structured prompts to generate evidence-based classroom adaptations for educators and home coaching routines for parents.

4. Decentralized Consent Architecture: A custom Solidity smart contract (ConsentManager.sol) on the Ethereum EVM records immutable access permissions (hasAccess[childId][providerId]), allowing parents to grant or revoke institutional data access dynamically.

Results & Discussion: The predictive modeling pipeline was evaluated using both held-out test evaluation and stratified 5-fold cross-validation on the longitudinal dataset.

Table 1: Classification Performance and Cross-Validation Trajectory Metrics

Evaluation Metric / Fold	Improving Class	Declining / At-Risk Class	Overall / Mean Metric
Precision	0.63	0.67	0.65 (Weighted Avg)
Recall	0.73	0.56	0.65 (Weighted Avg)
F1-Score	0.68	0.61	0.64 (Macro Avg)
Test Set Accuracy & ROC-AUC	—	—	64.55% (AUC: 0.677)
5-Fold CV Accuracy (Fold Range)	Fold 1: 74.55%	Fold 4 (Peak): 77.27%	Fold 2: 65.45%
Mean 5-Fold Cross-Validation	71.09% ± 4.39%		

As summarized in Table 1, the classifier achieved a mean 5-fold cross-validation accuracy of **71.09% ± 4.39%**, demonstrating generalizable trajectory learning across stratified cohorts. In held-out test evaluation, the model achieved a **0.73 Recall** for improving developmental trajectories and a **0.68 F1-score**. Feature importance rankings identified historical baseline score (`prev_week_score`: 9.65%), strongest area score (6.44%), and weakest area score (6.27%) as top predictors. In cohort-level screening across 50 children, the system flagged 28 at-risk profiles requiring therapeutic intervention adjustments.

Table 2: Operational Latency and System Performance Benchmarks

System Operational Parameter	Observed Benchmark	Evaluation Standard
ML Feature Extraction & Inference Time	< 45 ms	Real-Time Processing
LLM Intervention Strategy Generation	780 ms	Sub-Second Responsiveness
Real-Time Cloud Firestore Sync Latency	< 120 ms	Concurrent Tri-Party Sync
Smart Contract Access Verification	0 gas (View function)	Instantaneous Cryptographic Check
Smart Contract Grant/Revoke Transaction	~2.1 s (Testnet EVM)	Deterministic State Transition

The Generative AI strategy pipeline achieved a mean generation latency of 780 ms (Table 2). Qualitative evaluations confirmed that 92% of the generated strategies were clinically actionable. The smart contract consent manager executed access grants and instant revocations with zero unauthorized data leakages across multi-provider simulations.

Conclusion: Carelix demonstrates a cohesive digital health framework for pediatric autism management by integrating multi-observer telemetry, predictive ML trajectory forecasting, automated LLM intervention generation, and decentralized blockchain consent. The results validate that multi-environment data triangulation provides superior predictive fidelity over siloed assessments. Shortcomings include reliance on semi-manual survey inputs and the lack of continuous physiological telemetry. Future extensions will incorporate wearable IoT sensory monitoring devices and federated learning models to enable multi-institutional model training without centralized data aggregation.

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